



INTEGRATED WASTE MANAGEMENT SUITE



A unified digital platform for waste management.

Overview of IWMS

The Integrated Waste Management Suite (IWMS) is a digital platform built to modernize, streamline, and optimize municipal solid waste management. It brings together IoT-enabled sensors, GPS-tracked vehicles, QR-based household verification, CCTV monitoring, and mobile/web platforms to ensure seamless operations.

The system ensures:

- Transparency & Accountability – End-to-end traceability of waste movement.
- Fraud Prevention – Eliminates manual loopholes with automated verification.
- Automated Collection Verification – QR codes validate household and bin-level pickups.
- Accurate Weighment Tracking – IoT load cells and weighbridge integration.
- Real-time Citizen Engagement – Notifications, feedback collection, and grievance handling via app/portal.
- Regulatory Compliance – Aligned with Solid Waste Management (SWM) Rules for sustainable urban operations.
- Sustainability – Promotes segregation, recycling, composting, and eco-friendly disposal.
- Processing – Scientific segregation and treatment of wet, dry, hazardous, sanitary, and glass waste.
- Disposal & Rejects – Safe disposal of non-combustible, non-compostable, non-reusable, non-recyclable rejects at engineered landfills.
- Landfill Management – Tracks inflow, balance life, fire safety, informal activity monitoring, compliance.

IWMS Model

The system integrates web and app-based monitoring tools with an IoT-enabled navigation system to ensure efficient waste collection. It uses QR code-based tracking for accountability, connects with the Integrated Command and Control Centre (ICCC) for centralized oversight, and includes a grievance redressal system to address citizen concerns effectively. All operations will strictly follow the Solid Waste Management (SWM) Rules to ensure compliance with regulatory protocols.

The system ensures:



Web & App Based Monitoring tools



IoT Based Navigation System



Sensor based waste collection



ICCC (Integrated command and control centre)



Grievance Redressal system

Functional Modules of IWMS

1. Collection & Verification

- Ensures waste pickup validation using QR codes, event logging, and citizen notifications.

2. D2D Collection & Logistics Management

- Monitors vehicles in real-time with GPS, route optimization, fuel tracking, and automated fraud prevention.

3. Resource Management

- Digitally manages staff attendance, route assignments, and performance reports.

7. Grievance Redressal Management

- Captures citizen complaints and feedback with escalation workflows and analytics insights.

4. Weighbridge Management

- Tracks waste quantities via IoT sensors, weighbridges, and reconciliation with alerts.

5. Waste Management

- Logs and manages waste segregation and processing across multiple treatment facilities.

6. Landfill Management

- Ensures safe, transparent, and compliant operations through waste tracking, analytics, fire safety, activity monitoring, and regulatory reporting.



Collection & Verification



Collection & Verification

QR Tagging for Verification

- QR codes for households, bins, gated communities, apartments, industries, and institutions.

Pickup Event Logging

- Records GPS location, timestamp, waste image, and weighment
- Creates tamper-proof digital logs
- Live Tracking and Travel time per zone

Exception Handling

- Flags cases like locked houses, no waste, or damaged bins
- Enables reporting and corrective actions

Resident Notifications

- SMS/App alerts 15 minutes before collection vehicle arrival
- Improves citizen participation and readiness

Body Cameras for Operators

- Enhances transparency, safety, and fraud prevention
- Provides dispute resolution evidence

Asset Management:

- Capex – Vehicles, Bins, Equipment
- Opex – Fuel, Maintenance, Staff
- Attendance – Daily workforce tracking

D2D Collection & Logistics Management



D2D Collection & Logistics Management

GPS-Enabled Live Tracking

- Real-time visibility of all vehicles on a map
- Geo-fencing to prevent unauthorized movement

Route Planning & Monitoring

- Optimized route allocation
- Adherence tracking with alerts for deviations

Diesel Consumption Monitoring

- Tracks fuel usage & efficiency
- Prevents misuse & ensures cost control

Vehicle Capacity & Utilization Reports

- Monitors load vs. actual capacity
- Helps in efficient fleet utilization

Ring Fencing & Fraud Prevention

- Auto-detection of vehicle movement
- Vehicle photo & CCTV verification for accountability

Resource Management



Resource Management

Digital Attendance

- Photo, QR, or biometric-based attendance logging
- Prevents proxy or manual errors

Crew Assignment

- Allocate staff to specific routes and tasks
- Optimizes workforce utilization

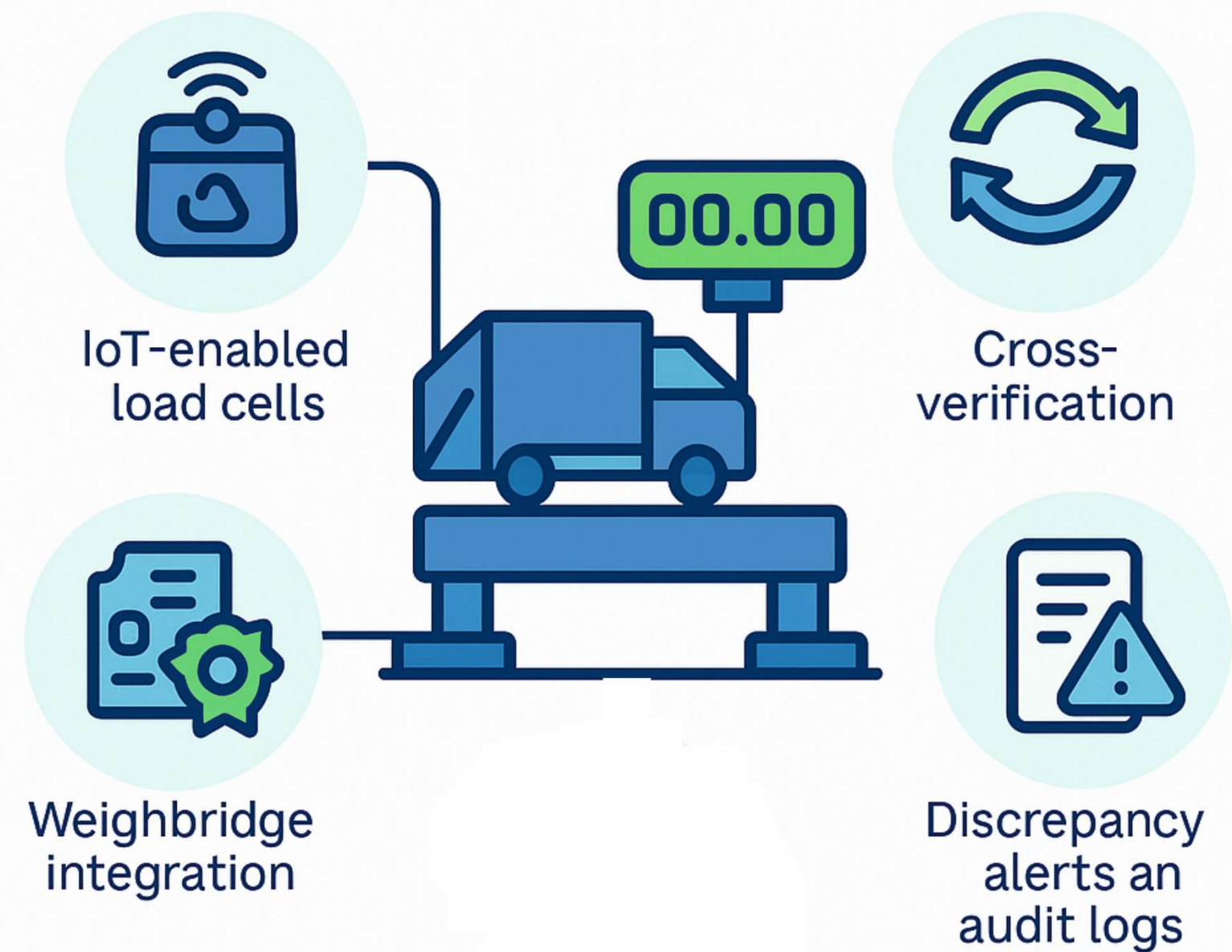
Performance Tracking

- Measure route coverage, number of stops, and citizen service ratings
- Identify top performers and gaps

Reports & Exceptions

- Generate daily and monthly attendance reports
- Highlight exceptions (absentees, missed pickups, late logins)

Weighbridge Management



Weighbridge Management

IoT-Enabled Load Cells

- Real-time monitoring of vehicle waste weights
- Accurate data collection and logging

Weighbridge Integration

- Transfer stations connected with digital weighbridges
- Seamless flow of weighment data

Cross-Verification

- Compares per-stop collected waste with vehicle's gross weighment
- Ensures consistency and prevents manipulation

Discrepancy Alerts & Audit Logs

- Instant notifications for mismatched weights
- Complete digital audit trail for accountability

Calibration Monitoring

- Alerts for calibration schedules
- Certificate renewal reminders for weighbridge compliance

Waste Management



Waste Management

Segregation Logging

- Tracks wet, dry, hazardous, sanitary, and glass waste categories

Processing Routes

- Organic → Composting / Biomethanization
- Dry → MRF (Material Recovery Facility) / Pyrolysis
- Mixed → Segregation & Composting

Disposal & Reconciliation

- Records disposal flow
- Aggregates data for reconciliation and compliance reports

Rejects (Definition & Handling under SWM Rules)

- Non-combustible, non-compostable, non-reusable, non-recyclable items

Examples: Broken commodes, construction debris, inerts, ceramic waste

Handling Protocol:

- Diverted to scientifically managed landfills or designated low-lying areas
- Ensures no contamination of recyclable and compostable streams

Landfill Management



Landfill Management

Waste Inflow Tracking

- Monitors all waste reaching the landfill, including processing rejects and mixed/unsegregated waste
- Integrated with weighbridge data

Analytics Dashboards

- Visualizes inflow trends, site balance life, forecast capacity, and diversion ratios
- Data-driven planning for landfill operations

Fire & Safety Management

- Equipped with surveillance cameras, thermal sensors, and suppression systems.
- Monitors hotspots and initiates alerts to prevent accidents and landfill fires.

Informal Activity Monitoring

- Detects rag pickers, animals, birds, or unsafe activities at landfill sites.

- Uses CCTV + AI-based analytics integrated with drone surveys for aerial monitoring.
- Helps identify unauthorized dumping and unsafe human activities.

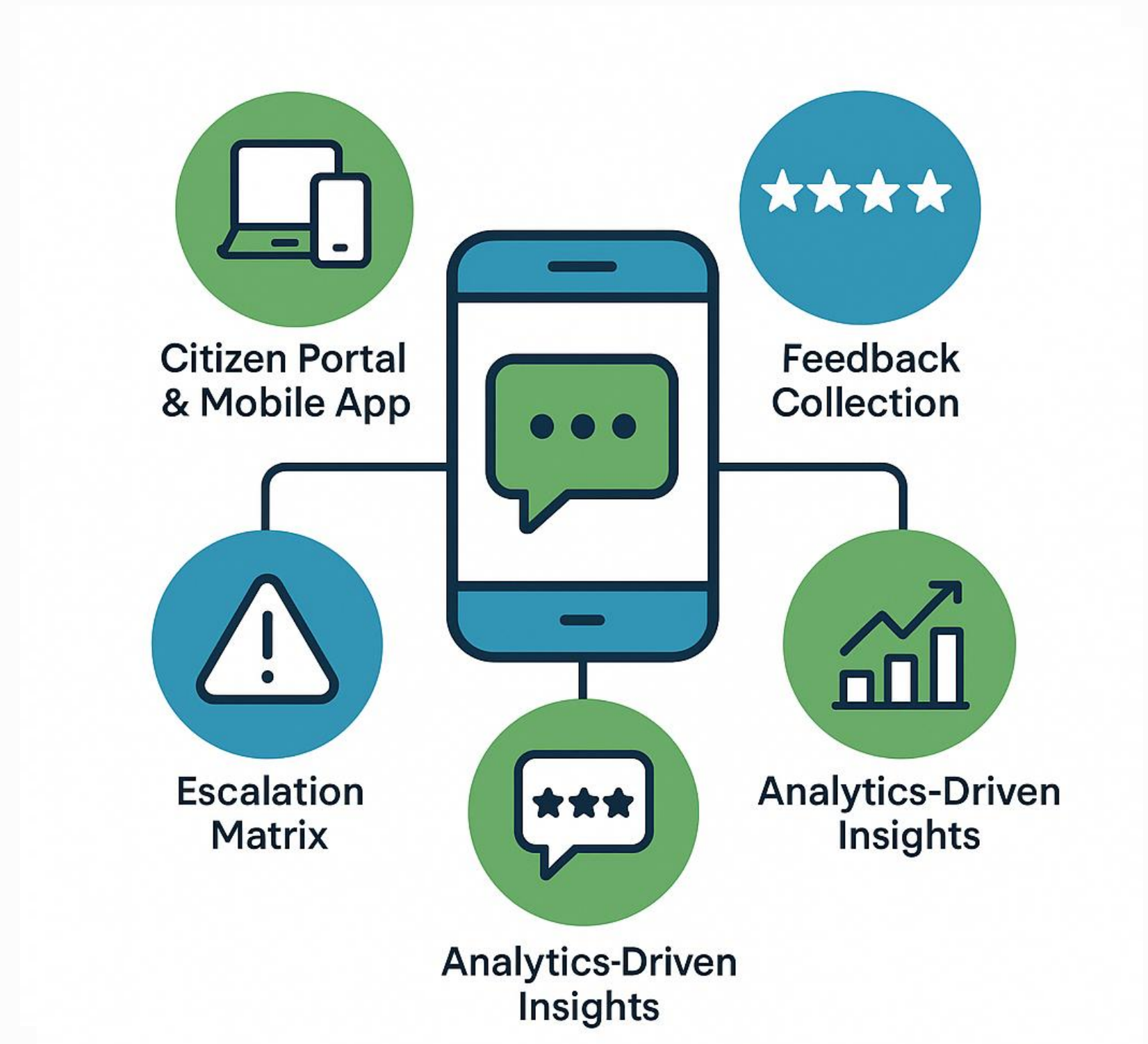
Compliance & Reporting

- Ensures adherence to SWM Rules, environmental norms, and safety standards
- Generates structured reports for authorities

Landfill Capacity

- Tracks how much waste a landfill can safely hold before closure.
- Calculated from engineered volume, compaction, and cover requirements, updated with weighbridge, drone/GIS, and 3D modeling.
- Dashboards show current utilization, balance life, and forecast scenarios, enabling smart planning for new cells, expansion, or diversion.

Grievance Redressal Management



Grievance Redressal Management

Citizen Portal & Mobile App

- Easy access for citizens to raise complaints anytime
- Transparent communication channel

Feedback Collection

- Ratings (1–5 stars) after each waste pickup
- Helps monitor service quality

Escalation Matrix

- Automatic routing of unresolved complaints to higher authorities
- Ensures accountability and quick resolution

Analytics-Driven Insights

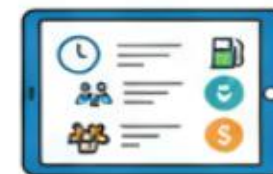
- Tracks complaint trends, response times, and feedback patterns
- Data-driven service improvements

Benefits of IWMS

- Focus on **Reduce, Reuse, Recycle**
- **Accountability:** Live data (time, staff, fuel, costs)
- **Efficiency:** End-to-end waste collection with checks & balances
- **Transparency:** Prevents waste leakage & ensures compliance



Accountability



Live data (time, staff, fuel, costs)

Efficiency



End-to-end waste collection with checks & balances

Transparency



Prevents waste leakage & ensures compliance

Key Technologies

- **GPS Tracking:** Route optimization, diesel usage, unauthorized movement
- **QR Codes:** Pickup verification at household/bin level
- **Smart Bin Sensors:** Fill-level, weight monitoring, automated alerts
- **Thermal Cameras:** Fire safety & landfill hotspot detection
- **CCTV + AI:** Drone/GIS monitoring, informal activity tracking

GPS Tracking



Route optimization,
diesel usage,
unauthorized movement

QR Codes



Pickup verification at
household/bin level

Efficiency



Fill-level, weight
monitoring,
automated alerts

GPS Tracking



Live data time)
staff, fuel, costs

Smart Bin Sensors



Fire safety & landfill
hotspots
detected alerts

CCTV + AI



Drone/GIS monitoring,
informal activity
tracking

Citizen Interface & Governance

- **Help Desk:** Handles complaints, missed pickups
- **Grievance Redressal:** Escalation workflows & analytics
- **Citizen Ratings:** AI-based ratings on segregation & compliance
- **Fines & Enforcement:** Penalties for violators, repeat offenders
- **Visitor Management System:** Track external entries

Help Desk



Handles complaints,
missed pickups

Grievance Redressal



Escalation workflows
& analytics

Citizen Ratings



AI-based ratings on
segregation &
compliance

Fines & Enforcement



Penalties for violators,
repeat offenders

Visitor Management System



Track external entries

CCTV + AI



Track external entries

Compliance & Rules

- Aligned with SWM Rules 2016
- User charges & fines for mixed waste
- Monitoring waste segregation at source
- Automated reporting for transparency & accountability

GPS Tracking



Route optimization,
diesel usage,
unauthorized movement

QR Codes



Pickup verification at
household/bin level

Efficiency



Fill-level, weight
monitoring,
automated alerts

GPS Tracking



Live data time)
staff, fuel, costs

Smart Bin Sensors



Fire safety & landfill
landfill hotspots
detected alerts

CCTV + AI



Drone/GIS monitoring,
informal activity
tracking

Financial Sustainability

- **User Charges:** ₹5–₹1000 per month (household to bulk users)
- **Revenue from Fines:** For non-compliance & violations
- **Government Subsidies:** Upfront spend → later reimbursement



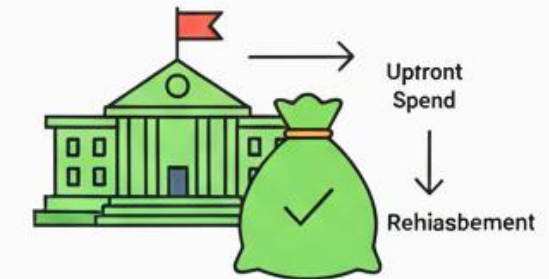
User Charges

₹5- ₹1000 per month
(household to bulk users)



Revenue from Fines

For non-compliance
& violations



Government Subsidies

Upfront spend → later
reimbursement

Phased Implementation (12-Month Roadmap)

Phase 1 (0–1 month): Weighbridge Monitoring System (accurate inflow measurement, integrated with landfill and processing data)

Phase 2 (Month 1–2): Processing System Setup (facility mapping, segregation of organic, dry, and mixed waste)

Phase 3 (Month 2–3): Landfill Management Module (inflow tracking, analytics dashboards, landfill capacity, fire & safety, compliance)

Phase 4 (Month 3–6):

- ✓ GPS Tracking & Route Optimization
- ✓ QR Code Rollout for households, apartments, bulk waste generators, and bins

Phase 5 (Month 6–9): User Charge (integrated billing, online/offline payment options)

Phase 6 (Month 9–12): Grievance Redressal Mechanism (citizen app/portal, ticketing, escalation matrix) .

Outcome: Full IWMS operational in 12 months, with scalability for faster rollout if manpower is available.

THANK YOU

We can make this vision turns to reality through innovative thinking and collective action.

FOR MORE INFORMATION

 www.zigma.in

 Chennai
Singapore

